Generative AI Can Harm Teaching

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University of Pennsylvania

INFORMS Annual Meeting 2026

Teachers already use generative AI

60%

of U.S. K-12 teachers used an AI tool last school year

Gallup & Walton Family Foundation 2025

1/3

use it every week

Gallup & Walton Family Foundation 2025

No causal evidence on its effects on students.

AI could increase or decrease the quality of teaching

Optimistic

High-quality content.

Time freed for rehearsal, relationships and differentiation.

Oreopoulos et al. 2024

Pessimistic

Teachers offload the pedagogical thinking.

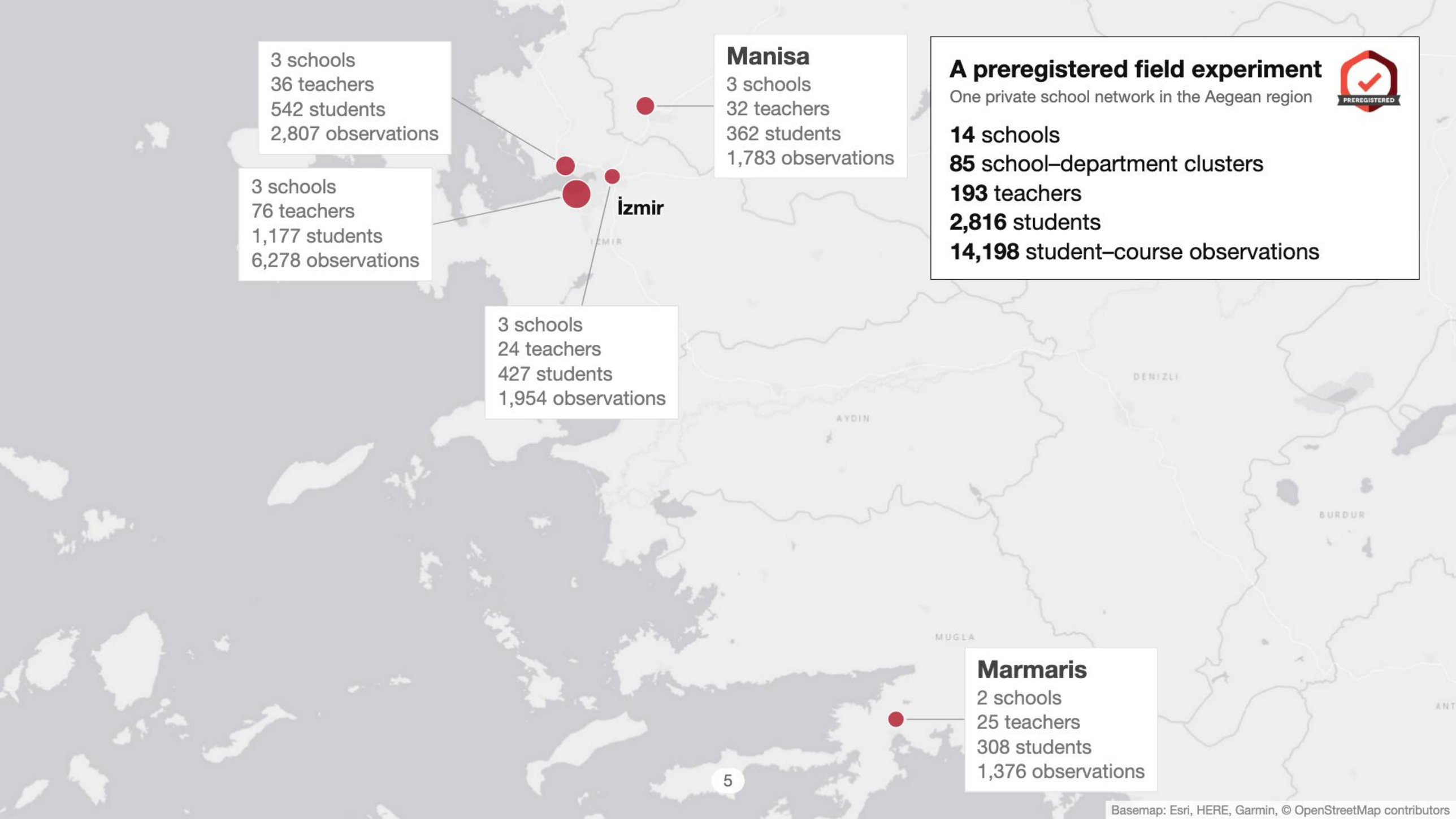
Materials come out polished but generic.

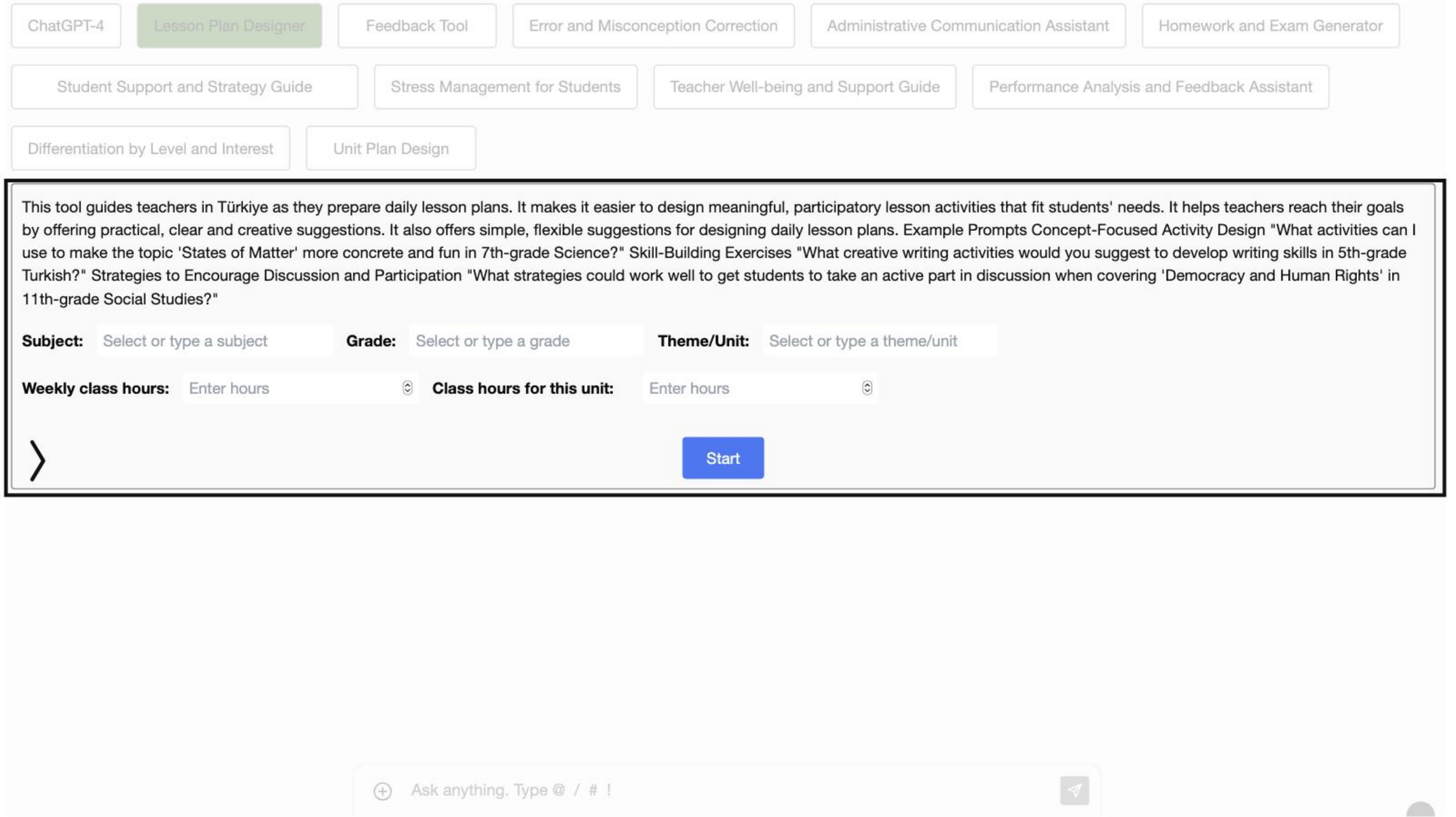
Time saved might not be reinvested.

Doshi & Hauser 2024
Farber et al. 2014

Five questions about what happens when teachers use AI

1. Do teachers use the tool?
2. What do they use it for?
3. How do students feel about their classes?
4. Do students learn more or less?
5. What do teachers come to believe about AI?


3 schools
36 teachers
542 students
2,807 observations
Manisa
3 schools
32 teachers
362 students
1,783 observations
3 schools
76 teachers
1,177 students
6,278 observations
İzmir
3 schools
24 teachers
427 students
1,954 observations
A preregistered field experiment
One private school network in the Aegean region
PREREGISTERED
14 schools
85 school–department clusters
193 teachers
2,816 students
14,198 student–course observations
İZMİR
AYDIN
DENİZLİ
BURDUR
MUĞLA
Marmaris
2 schools
25 teachers
308 students
1,376 observations
Basemap: Esri, HERE, Garmin, © OpenStreetMap contributors


ChatGPT-4
Lesson Plan Designer
Feedback Tool
Error and Misconception Correction
Administrative Communication Assistant
Homework and Exam Generator
Student Support and Strategy Guide
Stress Management for Students
Teacher Well-being and Support Guide
Performance Analysis and Feedback Assistant
Differentiation by Level and Interest
Unit Plan Design
This tool guides teachers in Türkiye as they prepare daily lesson plans. It makes it easier to design meaningful, participatory lesson activities that fit students' needs. It helps teachers reach their goals by offering practical, clear and creative suggestions. It also offers simple, flexible suggestions for designing daily lesson plans. Example Prompts Concept-Focused Activity Design "What activities can I use to make the topic 'States of Matter' more concrete and fun in 7th-grade Science?" Skill-Building Exercises "What creative writing activities would you suggest to develop writing skills in 5th-grade Turkish?" Strategies to Encourage Discussion and Participation "What strategies could work well to get students to take an active part in discussion when covering 'Democracy and Human Rights' in 11th-grade Social Studies?"
Subject:
Select or type a subject
Grade:
Select or type a grade
Theme/Unit:
Select or type a theme/unit
Weekly class hours:
Enter hours
Class hours for this unit:
Enter hours
Start
Ask anything. Type @ / # !



Control

Business-as-usual professional development



AI Access

A one-hour training and a custom GPT-4o tool built on the national curriculum



AI Access + Reminder

+ weekly emails showing each teacher's own usage, as a nudge

Both AI arms are **pooled** and compared with control.
Estimates are **intent-to-treat**.

Unit of randomization: the school–department

Timeline

August–December 2024

Pre-period

Baseline standardized test scores collected

January 2025

Training

Teachers receive AI training and complete questionnaires

February–May 2025

Intervention Period

Treated teachers use AI tool
Control teachers teach as usual

Standardized test scores collected

May 2025

Endline

Teachers and students complete questionnaires

Fall Semester 2024 | Winter Break | Spring Semester 2025

Five questions about what happens when teachers use AI

1 Do teachers use the tool?

2 What do they use it for?

3 How do students feel about their classes?

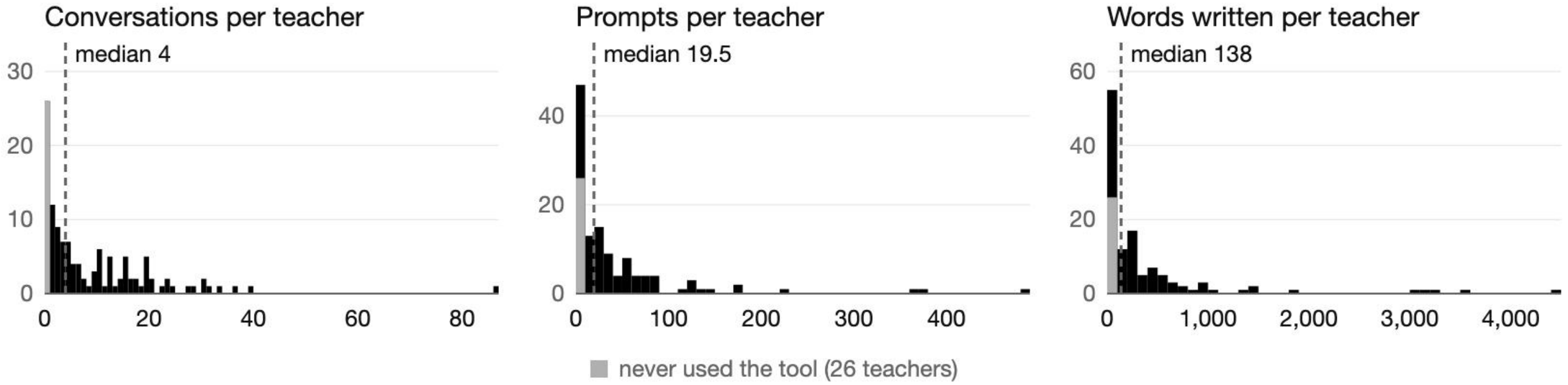
4 Do students learn more or less?

5 What do teachers come to believe about AI?

Most teachers did adopt the AI tool

78%

of treated teachers had at least one conversation



Five questions about what happens when teachers use AI

1 Do teachers use the tool?

2 What do they use it for?

3 How do students feel about their classes?

4 Do students learn more or less?

5 What do teachers come to believe about AI?

Teachers asked for finished materials



66%

of conversations asked for finished materials: lectures, homework, exams, syllabi

Can you make me a lesson plan on how atomic theory changed over time?
Course: Chemistry, 9th grade.

Here is a concise plan:
1. Objective: explain how atomic theories changed.
2. Activity: timeline discussion.
3. Assessment: short-answer quiz.
[…]

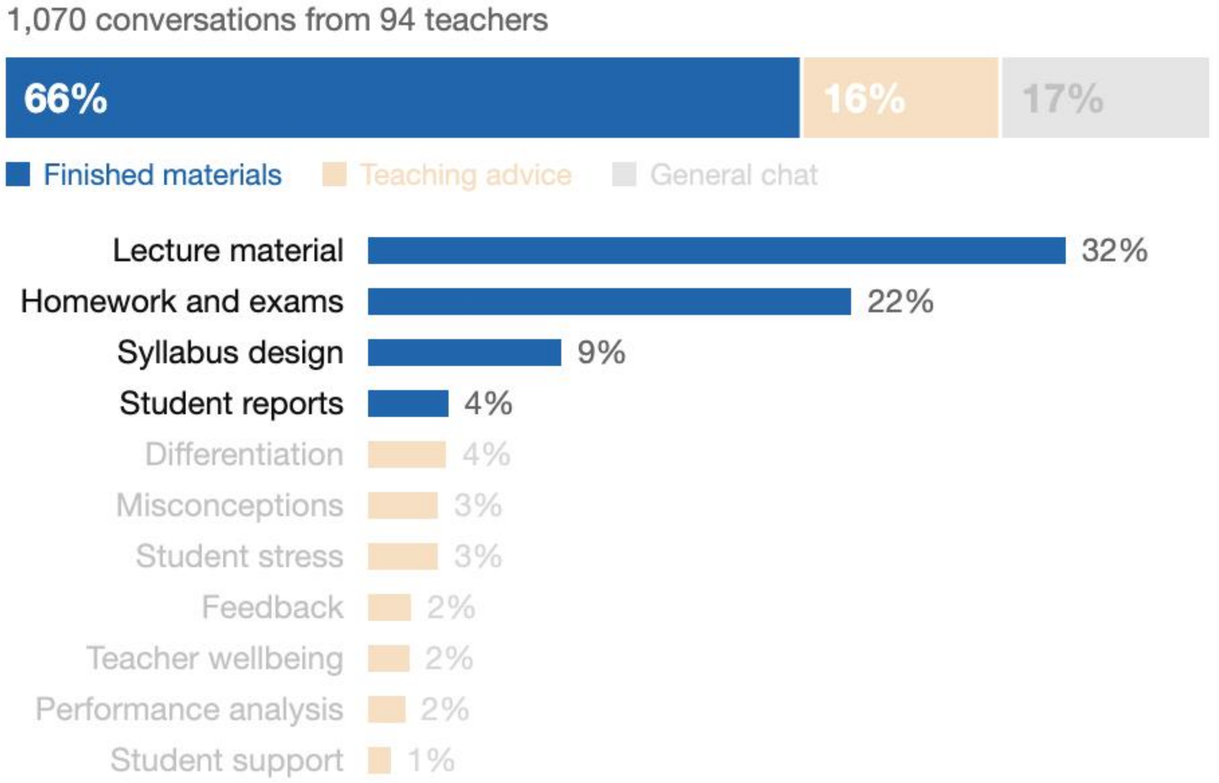
Thank you, that’s suitable.

You’re welcome.

Teachers asked for finished materials

66%

of conversations asked for finished materials: lectures, homework, exams, syllabi



Grade: [11] Unit: [Turkish agriculture and livestock]
Weekly hours: [4] Hours for the unit: [4]

Great, you will teach 11th graders about agriculture and livestock in Turkey. […] Are there errors or misunderstandings that students often have on this topic? […]

They make mistakes on where agricultural products are distributed on the map.

I see, they struggle with the distribution of agricultural products on the map. […] How are you teaching this topic right now? If you like, I can suggest activities or explanations that address these misunderstandings directly.

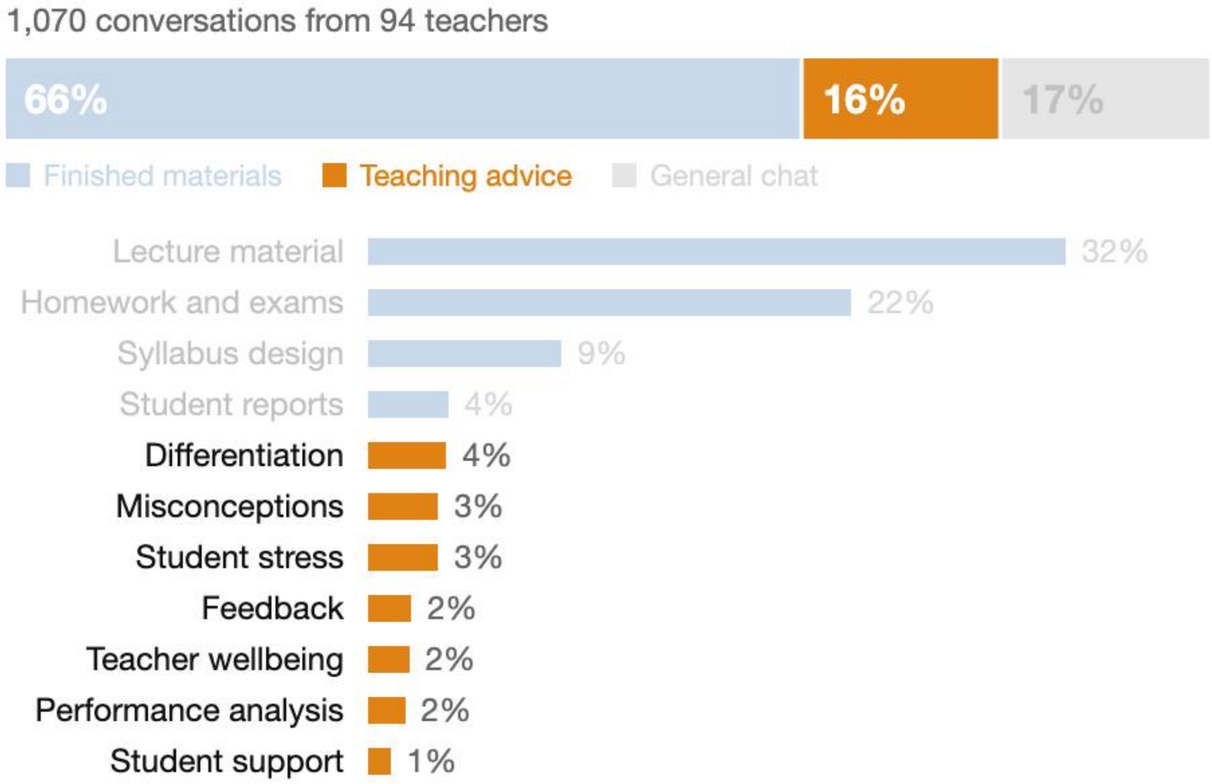
Suggest, please.

Of course! Here are a few suggestions: 1. Visual aids and map work […] 2. Group work and discussion […] 3. Analysis of wrong examples […]

Teachers asked for finished materials

16%

asked for teaching advice: differentiation, misconceptions, feedback



Five questions about what happens when teachers use AI

1 Do teachers use the tool?

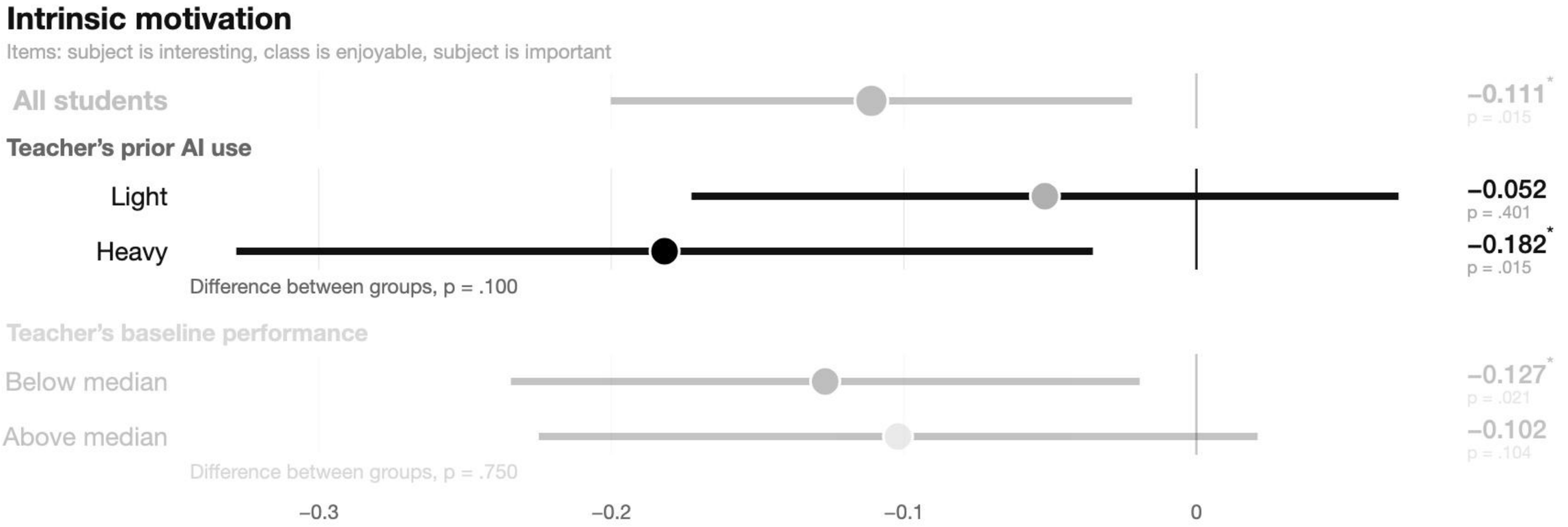
2 What do they use it for?

3 How do students feel about their classes?

4 Do students learn more or less?

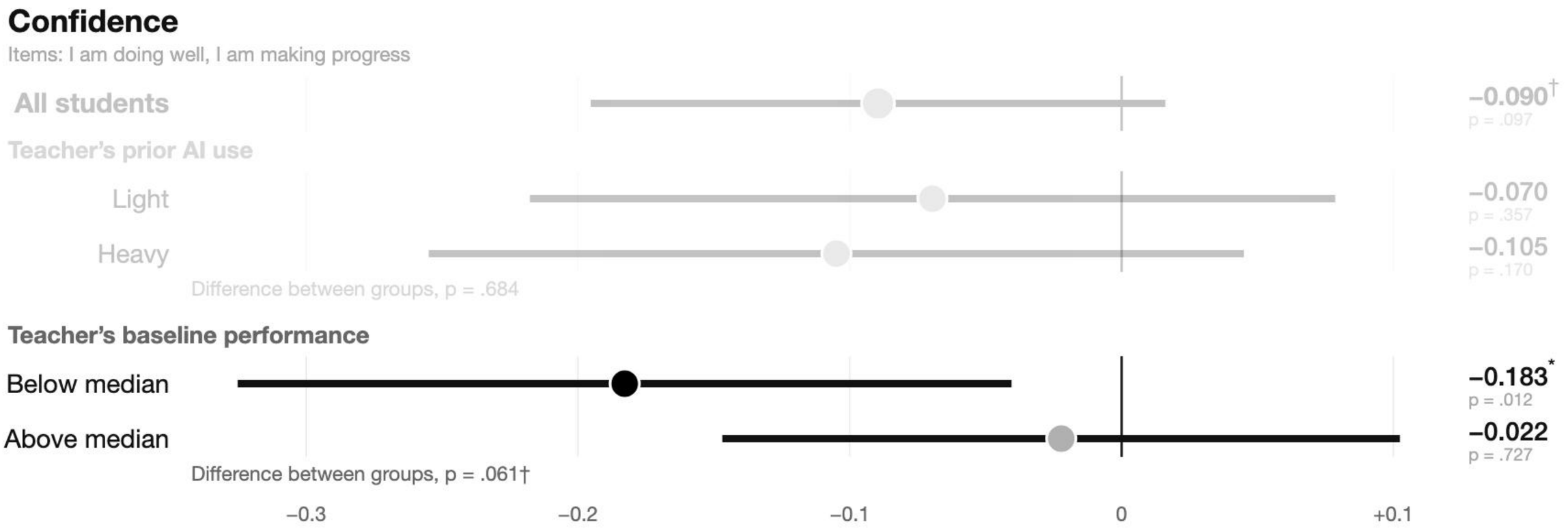
5 What do teachers come to believe about AI?

…especially for teachers higher in baseline AI use



Fredricks et al. 2004

Confidence fell less



Five questions about what happens when teachers use AI

1 Do teachers use the tool?

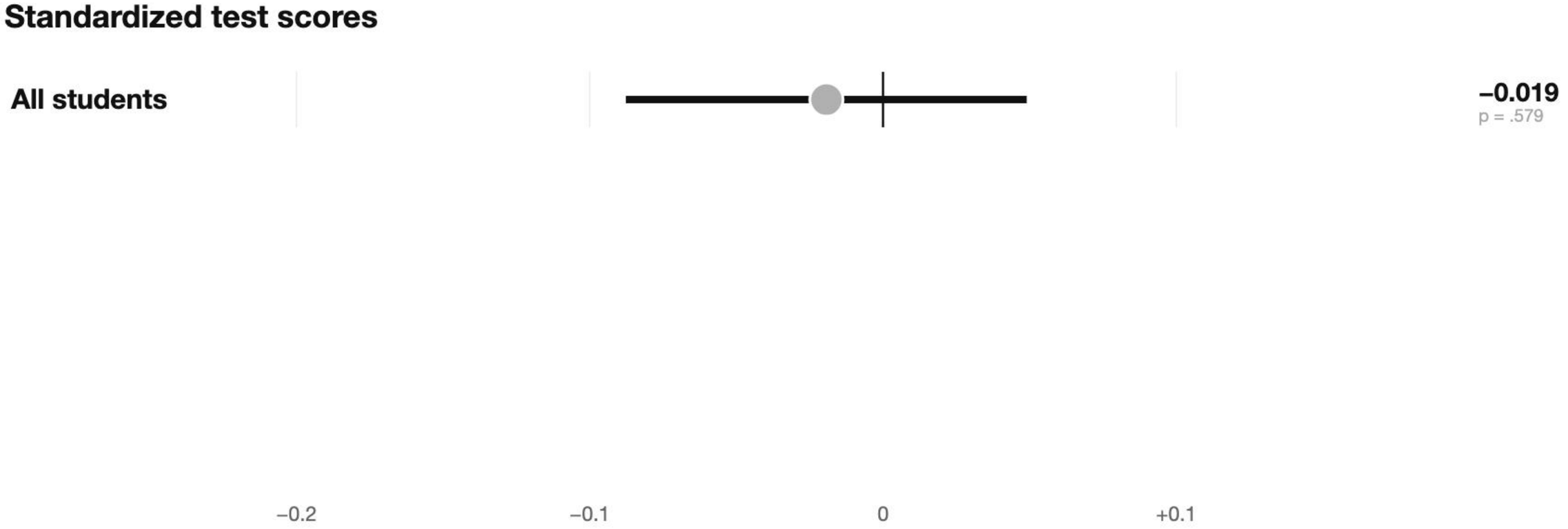
2 What do they use it for?

3 How do students feel about their classes?

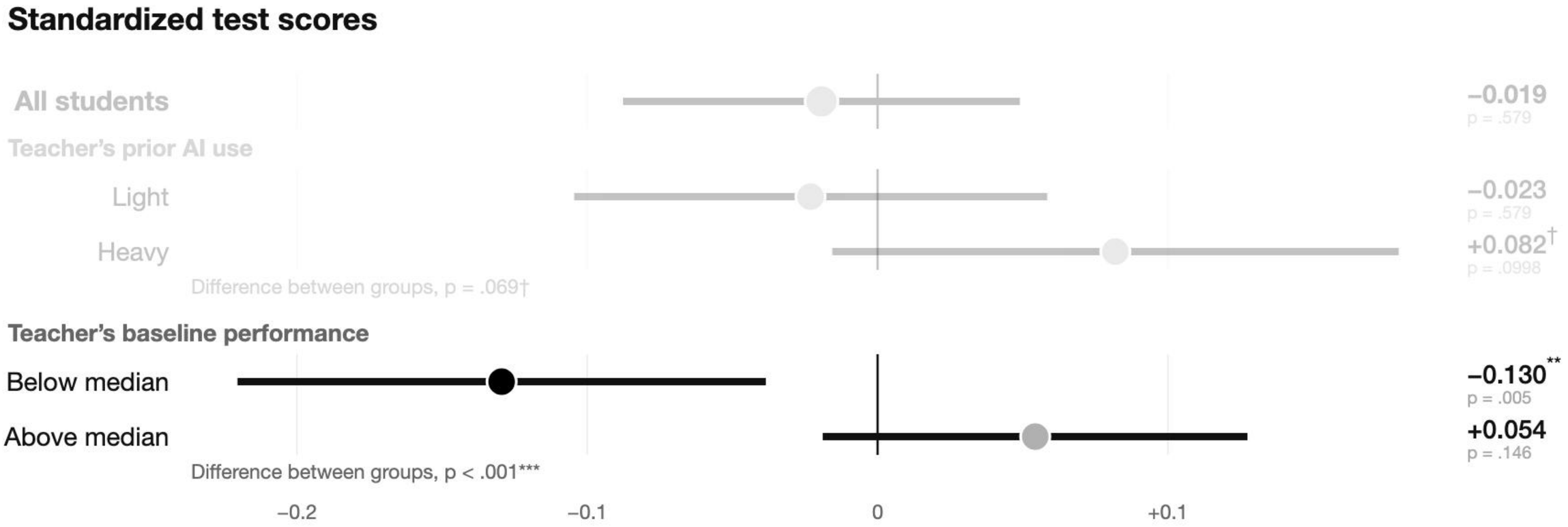
4 Do students learn more or less?

5 What do teachers come to believe about AI?

On average, AI did not move standardized test scores



…but it did harm students of lower-performing teachers



Five questions about what happens when teachers use AI

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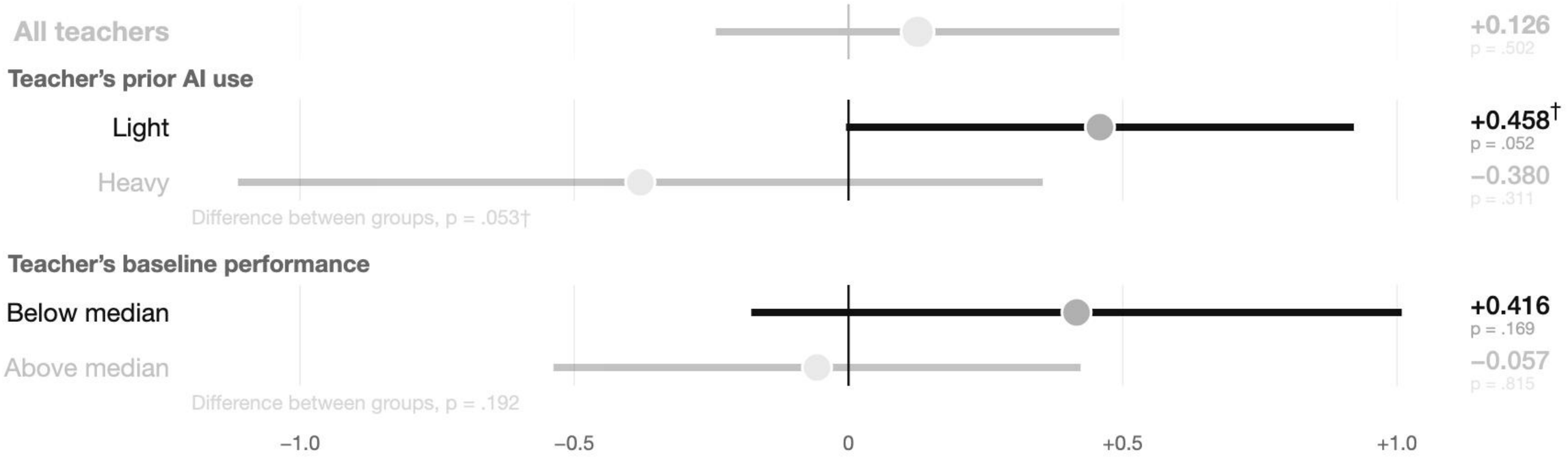
4 Do students learn more or less?

5 What do teachers come to believe about AI?

Teachers did not update their views on AI

even as their students' intrinsic motivation fell

"What overall impact do you believe AI tools have on students' learning outcomes?"



Why was AI harmful?

Effort signal

AI-made materials lack the teacher's own style, so students read them as fake or low-effort.

Doshi & Hauser 2024
Moon et al. 2026

Skill substitution

Teachers offload the pedagogical reasoning that makes materials work, and their skills erode.

Otis et al. 2026
Dell'Acqua 2022

Summary of results

Most teachers adopted AI, mostly delegating rather than co-creating.

Their students found those classes less interesting, enjoyable, and important.

Confidence fell less, mostly among students of weaker teachers.

On average, standardized test scores did not change.

But students of lower-performing teachers had learning losses.

Teachers did not update their views on AI.

AI should complement teachers, not replace them

Thank you

Benjamin Lira, blira@upenn.edu

Paper

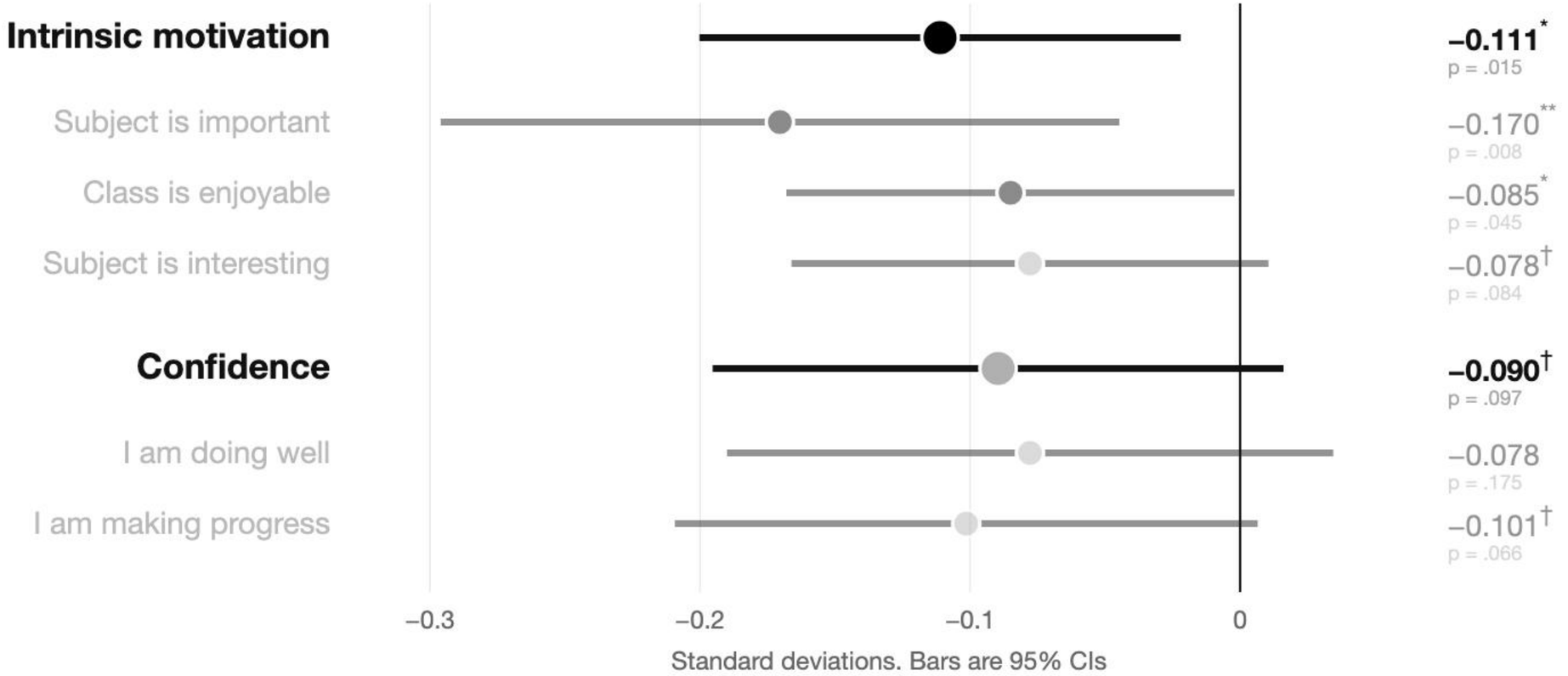
Code and data

Robustness checks

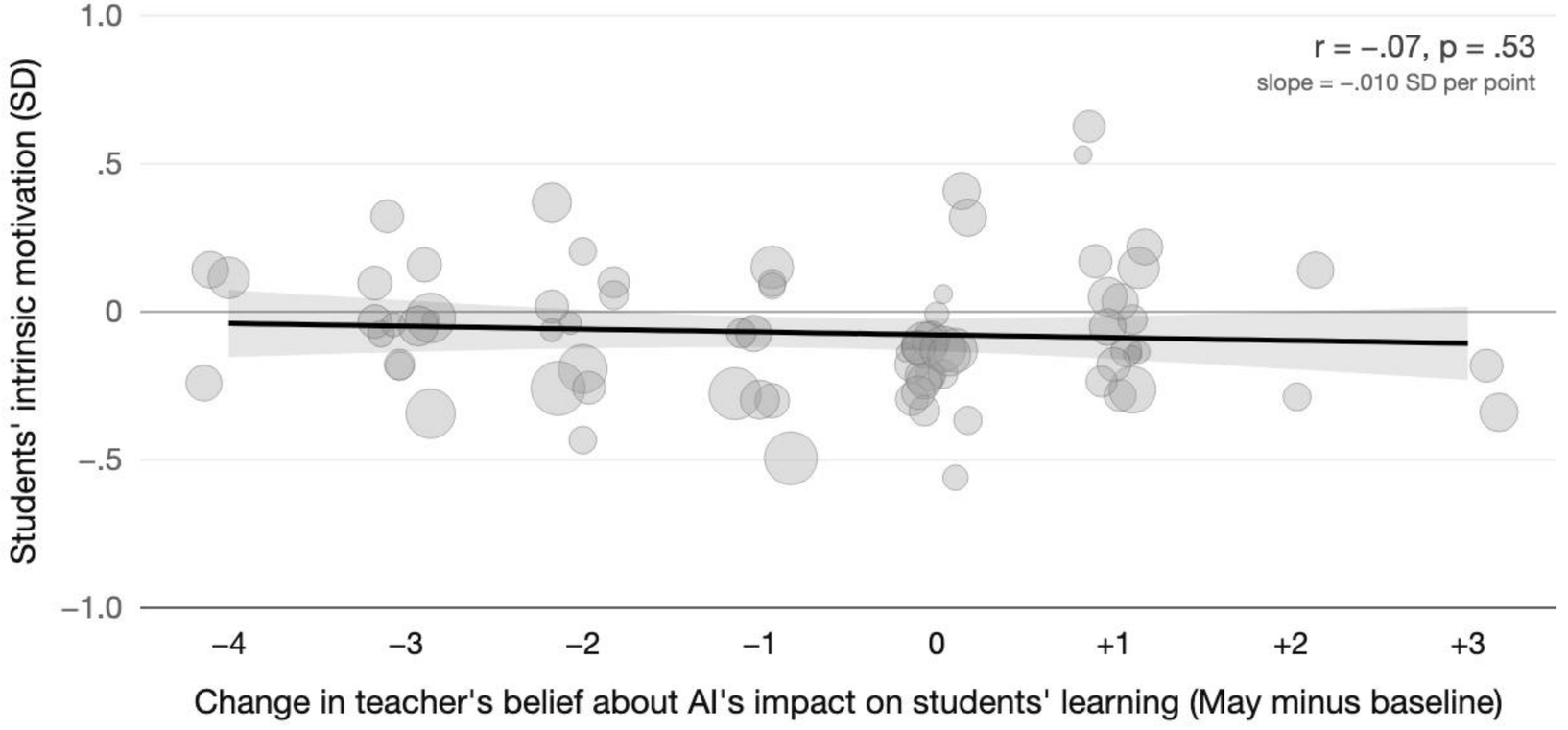
Attrition	Response rates did not differ by condition. Lee bounds stay negative.
Within-school variation	Every school had AI and control teachers.
Fixed effects	Still negative, though smaller, with department, school and student fixed effects.
Single items	No one survey item drives the result.
Arms and dosage	Same pattern for each AI arm and for AI-teacher share.

Lee (2009)

No single survey item drives the result

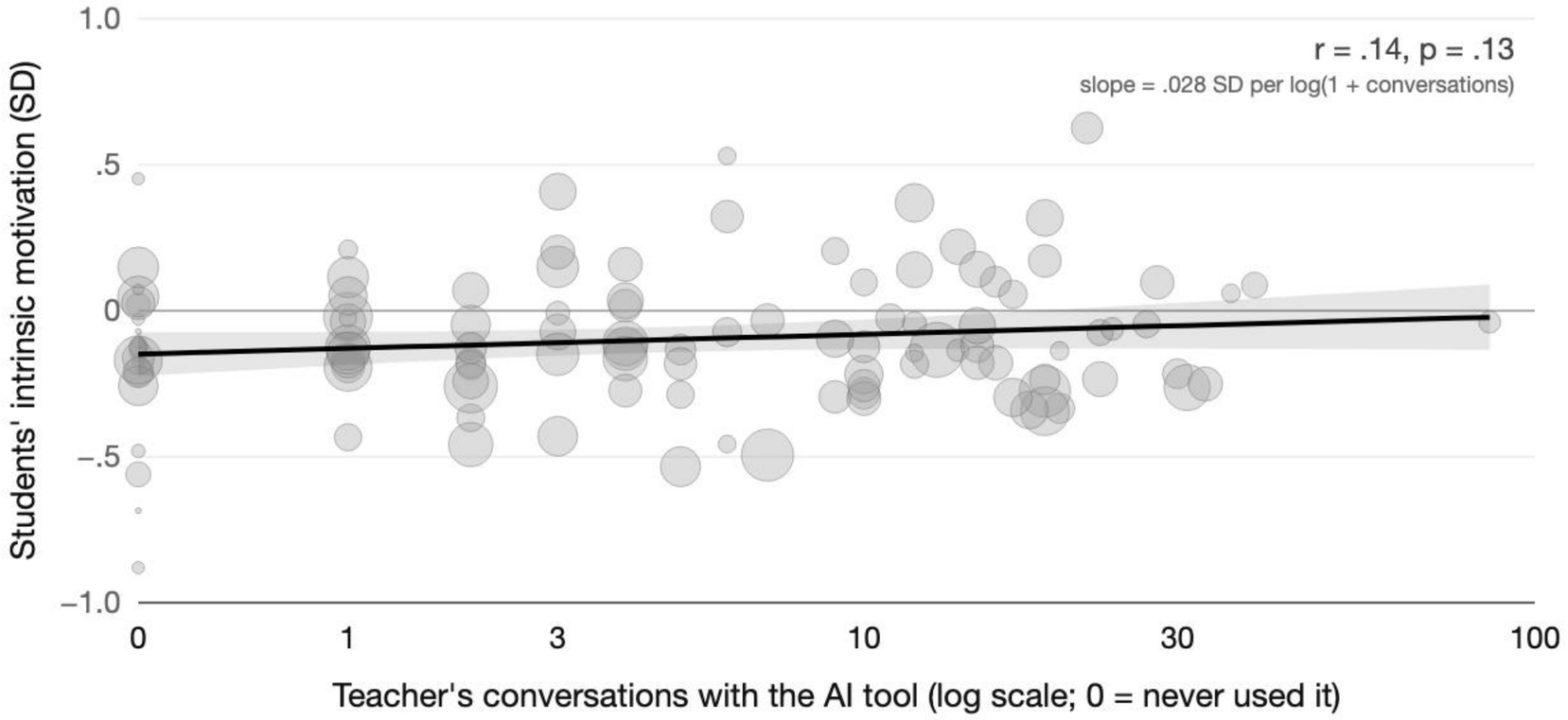


Teachers with less intrinsically motivated students did not grow more skeptical of AI



Descriptive; usage and beliefs were not randomized. N = 77 treated teachers. Dot area = number of surveyed students.

Students' intrinsic motivation did not fall with heavier AI use



Descriptive; usage and beliefs were not randomized. N = 114 treated teachers. Dot area = number of surveyed students.

Estimation

$$Y_{ic} = \alpha + \boldsymbol{\beta} \cdot \mathrm{AI}_d + \gamma \cdot Y_{ic}^{\mathrm{pre}} + \varepsilon_{ic}$$

i student (or teacher), *c* course, *d* school–department (unit of randomization)

β: effect of giving teachers AI (intent-to-treat, both AI arms pooled)

γ: baseline control for test scores and teacher beliefs (students had no pre-survey)

Standard errors clustered by school–department

$$Y_{ic} = \alpha + \beta_1 \cdot \mathrm{AI}_d + \beta_2 \cdot M + \boldsymbol{\beta_3} \cdot (\mathrm{AI}_d \times M) + \gamma \cdot Y_{ic}^{\mathrm{pre}} + \varepsilon_{ic}$$

β_3: does the effect differ by the teacher's prior AI use or baseline performance?
M = 1 if the teacher is above the median

The two AI arms look alike: adoption and usage

Outcome	AI Access	AI Access + Reminder	Arms differ (p)
Adoption Treated teachers only; no control comparison			
Used the tool at least once	80% 24/30	78% 70/90	1.00 means: .78
Conversations per teacher	median 8, mean 9.3	median 4, mean 8.8	.55 means: .83
Prompts per teacher	median 24.5, mean 31.1	median 17, mean 45.1	.96 means: .36
Words written per teacher	median 113.5, mean 241	median 144, mean 435.7	.80 means: .11
Usage Conversations of treated teachers			
Share finished materials	67% 185/278	66% 522/792	.93
Share teaching advice	18% 51/278	16% 125/792	.50
Prompts per conversation	median 2, mean 3.4	median 3, mean 5.1	.57 means: **.02**

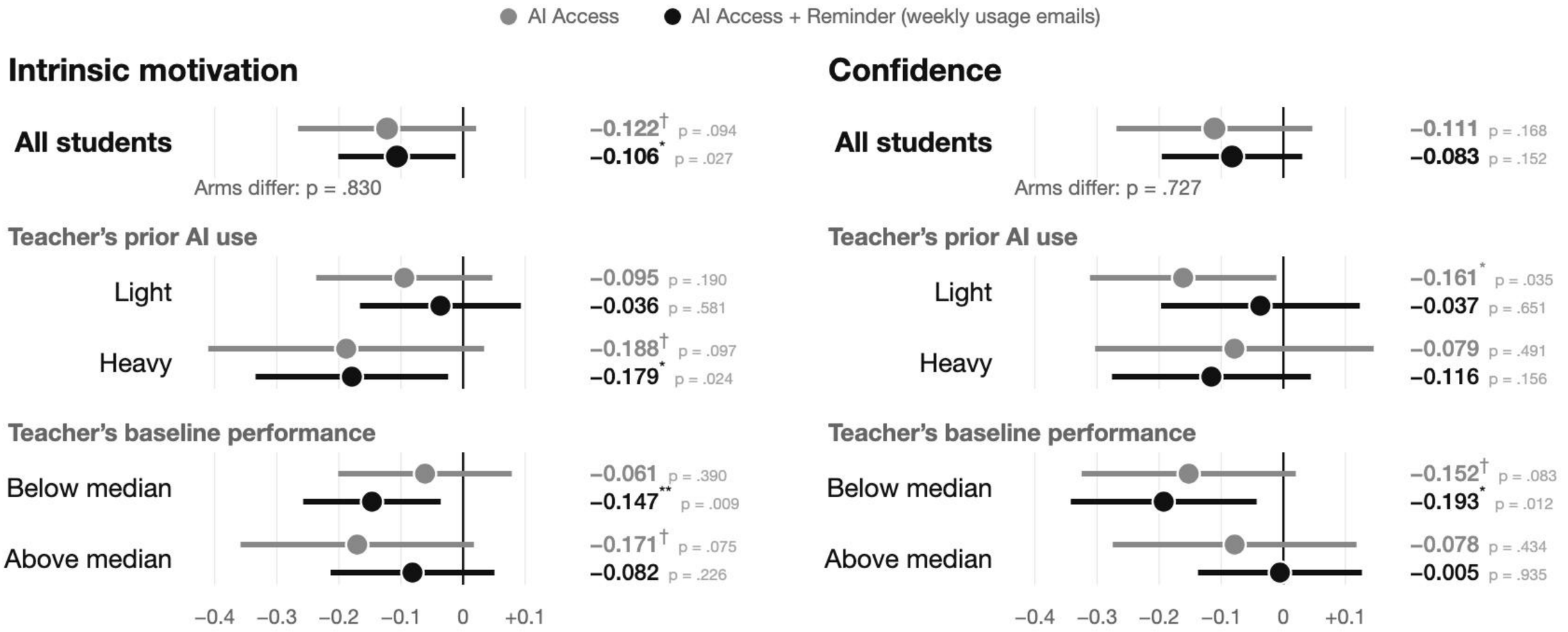
Treated teachers in the analytic sample: AI Access 30, AI Access + Reminder 90 (non-users count as zero); 278 and 792 conversations.
Used the tool: Fisher's exact test. Per-teacher counts: Mann–Whitney. Prompts per conversation: Mann–Whitney on teacher medians.
Means: OLS, SEs clustered by school–department (per teacher) or by teacher (per conversation). Shares: linear probability model clustered by teacher.

The two AI arms look alike: outcomes

Outcome	AI Access	AI Access + Reminder	Reminder – Access
Intrinsic motivation and confidence			
Intrinsic motivation	–0.12† p = .09 [–0.27, 0.02]	–0.11* p = .03 [–0.20, –0.01]	+0.02 [–0.13, 0.16] p = .83
Confidence	–0.11 p = .17 [–0.27, 0.05]	–0.08 p = .15 [–0.20, 0.03]	+0.03 [–0.13, 0.19] p = .73
Standardized test scores	–0.01 p = .81 [–0.09, 0.07]	–0.02 p = .54 [–0.10, 0.05]	–0.01 [–0.09, 0.06] p = .73
Teacher beliefs	+0.22 p = .38 [–0.27, 0.71]	+0.09 p = .66 [–0.30, 0.48]	–0.13 [–0.58, 0.32] p = .57

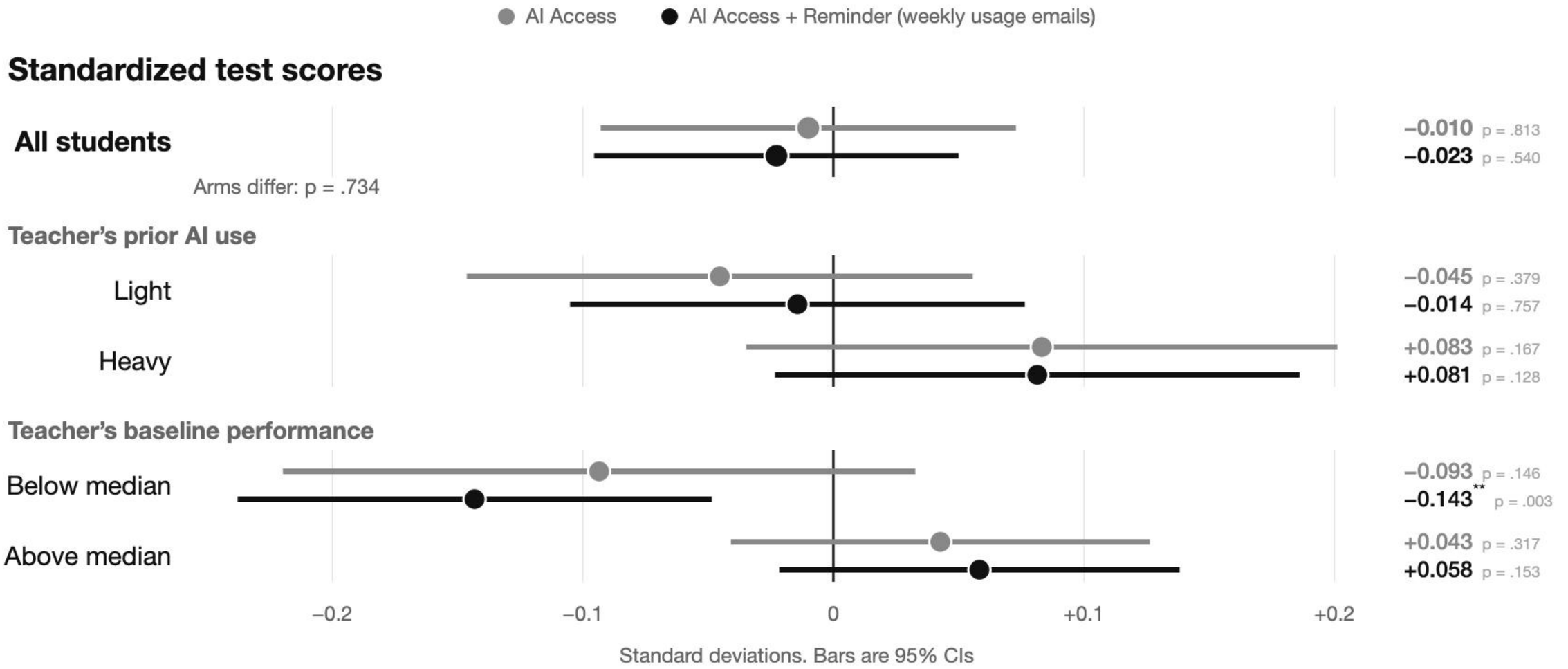
N, student–course survey responses: AI Access 1,808, AI Access + Reminder 5,754, control 3,730.
N, student–course test scores: AI Access 2,543, AI Access + Reminder 6,991, control 4,548. N, belief-survey teachers: AI Access 23, AI Access + Reminder 54, control 41.
Effects in SD vs control, 95% CI in brackets: OLS on both arm indicators, baseline control for test scores and beliefs.
Arm difference: Wald test of equal effects. SEs clustered by school–department.

By treatment arm: intrinsic motivation and confidence

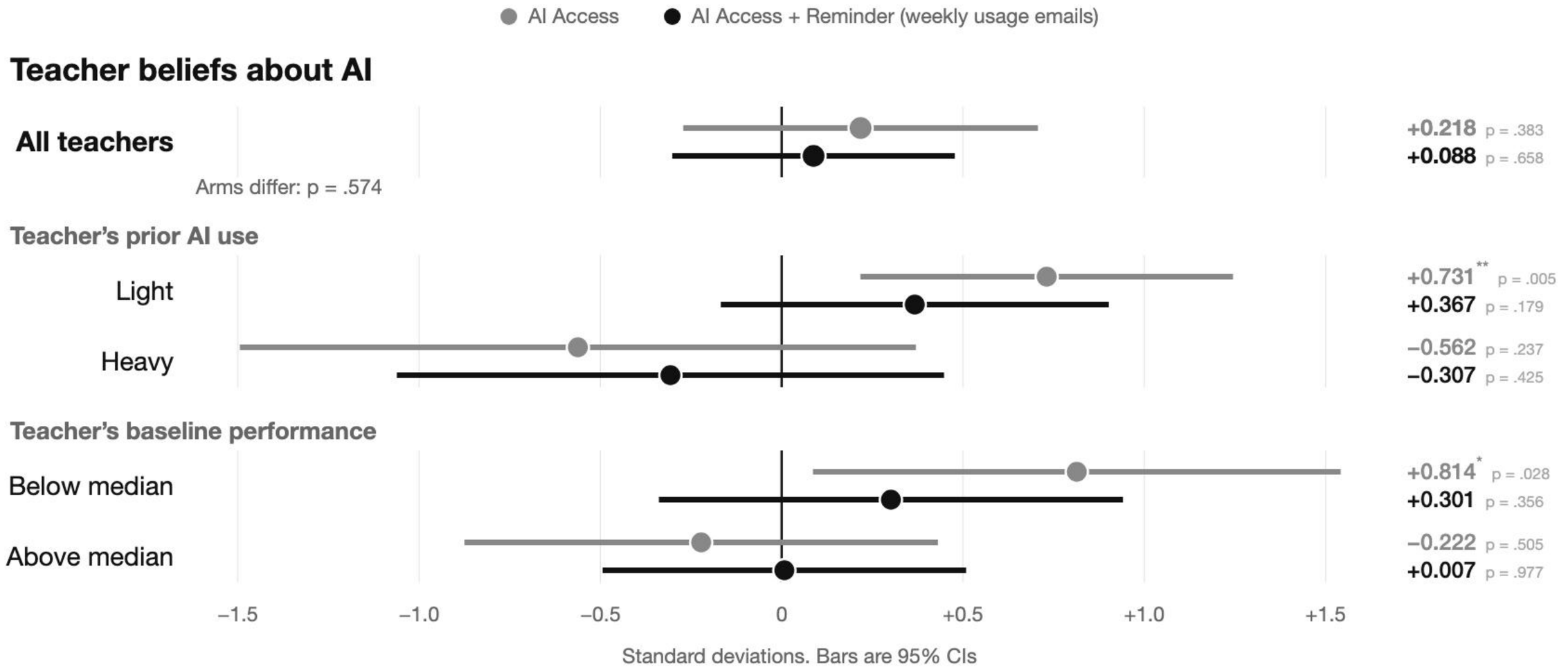


Standard deviations. Bars are 95% CIs

By treatment arm: standardized test scores



By treatment arm: teacher beliefs about AI



Limitations



Duration

We observe one semester.



Tool specificity

We test one AI tool design.



Contamination

Control teachers could use free AI tools.

A preregistered field experiment in 14 Turkish schools



14 schools, grades 5–12

85 school–department clusters

193 teachers

2,816 students

14,198 student–course observations

Overview

We randomly assigned teachers to get an AI assistant or teach as usual.
Most adopted it, mostly delegating rather than co-creating.
Their students found class less interesting, enjoyable, and important.
Standardized test scores held on average but fell for students
of weaker teachers. Teachers did not update their views on AI.

Your AI Usage Report This Week

AI Teaching Assistant team Sun, Feb 9, 2025

Hello Teacher,

Here is some data on your use of the AI tool:

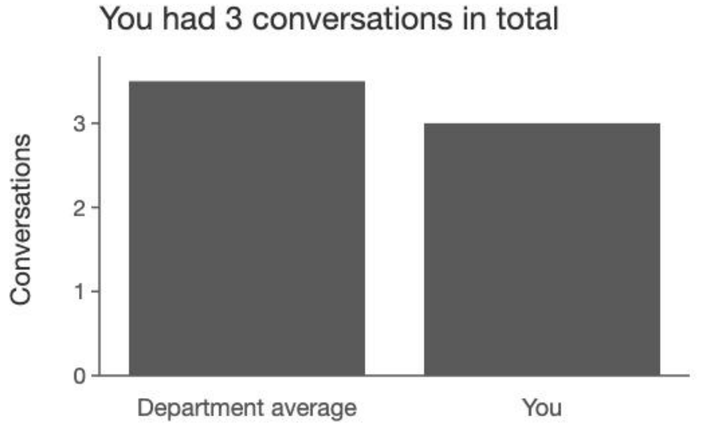
- You sent **8** messages.
- You had **3** conversations.
- You used the AI for about **1** minute in total.

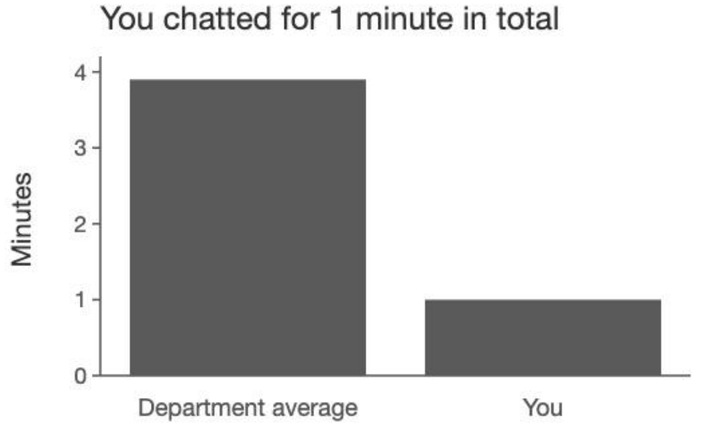
You've made a great start toward exploring more! Wishing you a successful new week.

Feel free to reach out with any questions.

Best wishes



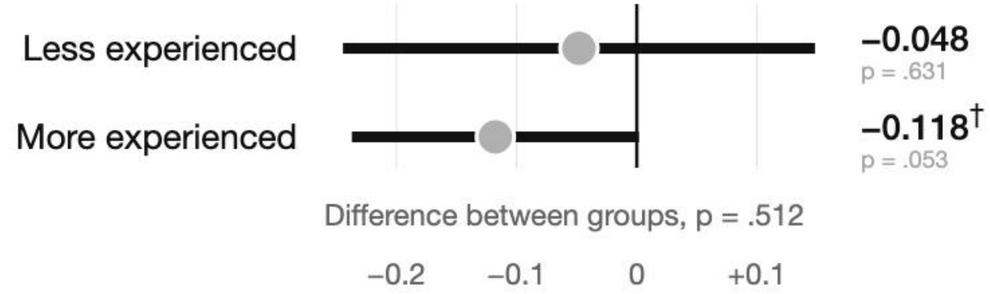




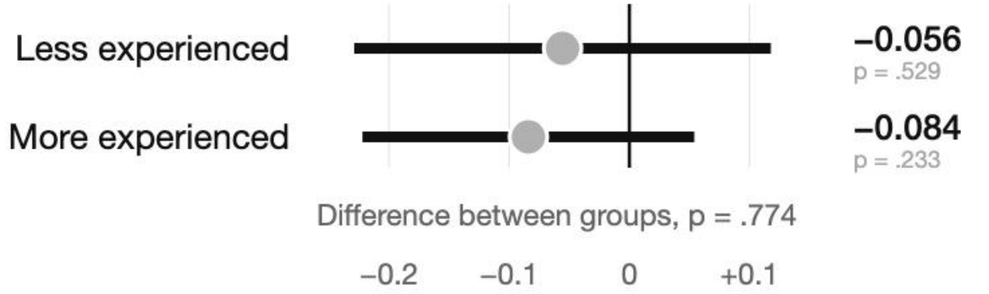
Weekly usage email to AI-arm teachers (Reminder arm). Original in Turkish.

Exploratory: by teacher experience

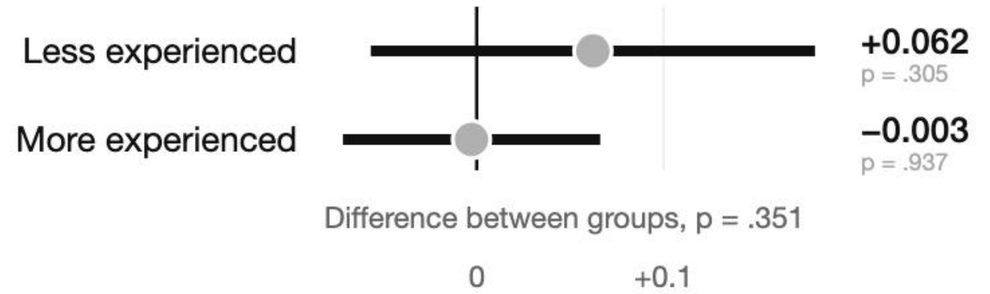
Intrinsic motivation



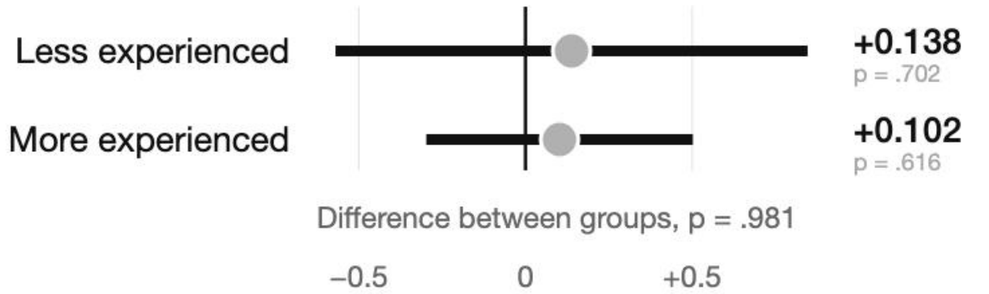
Confidence



Standardized test scores



Teacher beliefs about AI



Not preregistered. Experience is a 1–5 survey band (no year cut-points); groups: bands 1–3 vs 4–5; N teachers 36 vs 98.
65 of 199 teachers have no baseline survey and are left out. Standard deviations; bars are 95% CIs.